Sri Lanka Institute of Information Technology

AI-Based Social Integration and Inclusion Platform for University Freshers

Software Requirement Specification

IT4010 - Research Project

Project ID: 25-26J-154

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15/01/2026

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document specifies the software requirements for UniPulse, an AI-driven Social Integration and Inclusion Platform designed to support university freshers during their academic and social transition. This SRS covers the complete system, including its four core modules: Adaptive Student Profiling and Learning Persona Management, Social Inclusion Risk Detection and Early Intervention Management, Conversational Support and Student Wellbeing Management, and Intelligent Peer Matching and Relationship Evolution Management. This document serves as the definitive reference for developers, testers, project managers, and stakeholders involved in the design, implementation, validation, and deployment of the UniPulse system.

1.2 Document Conventions

This document follows specific conventions to ensure clarity and consistency. High-priority requirements are indicated with bold text, while figures, tables, and appendices are clearly labeled and referenced in the text. Functional and non-functional requirements are uniquely numbered to support traceability and management throughout the development lifecycle.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for developers and AI engineers responsible for implementing system modules and AI models, project managers who will plan and coordinate development, testers and QA engineers who will design test cases and validate functionality, university administrators and counselors who need to understand system capabilities and integration points, and ethics and compliance officers ensuring data privacy and ethical adherence. It is suggested that readers begin with the Introduction and Overall Description sections to gain a high-level understanding before proceeding with the detailed system features and external interface requirements.

1.4 Product Scope

UniPulse is an AI-based platform developed to address the critical mental health and social integration challenges faced by university freshers. The system leverages behavioral analytics and emotion recognition to proactively identify at-risk students and provide personalized interventions. By integrating four core components; adaptive profiling, peer matching, risk

detection, and conversational support, UniPulse aims to enhance student wellbeing, improve retention rates, and foster inclusive campus communities. In cases identified as critical, the platform will suggest a list of counselors to employees for follow-up, rather than providing students with direct counselor access. The platform is designed for scalability and cultural adaptation, with a focus on ethical data use and privacy preservation.

1.5 References

[1] E. F. T. J. L.-Z. O. Okan, "Mental health literacy and academic performance," Journal of Adolescent Health, vol. 68, no. 3, pp. 512-518, 2021.

[2] S. L. Ministry of Education, "Guidelines for Mental Health Promotion and Psychological Support in State Universities," Ministry of Education, 2022.

[3] G. Krüger, "How to Write a Software Requirements Specification (SRS) Document," Perforce, 11 Jul 2025. [Online]. Available: https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document. [Accessed 10 Jan 2026].

[4] E. C. A. N. W. E. G. Lattie, "Digital mental health interventions for depression, anxiety, and enhancement of psychological," Journal of Medical Internet Research, vol. 21, no. 7, p. e12869, 2019.

[5] A. Eser, "Student Loneliness Statistics," ZIPDO, 30 May 2025. [Online]. Available: https://zipdo.co/student-loneliness-statistics/. [Accessed 10 Jan 2026].

[6] S. B. S. K. A. A. Kumar, "Limitations of current wellbeing applications," Internet Interventions, vol. 24, p. 100381, 2021.

[7] NEWPORT INSTITUTE, "Depression in College Students: New Statistics and Research," NEWPORT INSTITUTE, 2022. [Online]. Available: https://www.newportinstitute.com/resources/mental-health/depression-on-college-campuses/?_gl=153k0sg_upMQ.._gaMTI1MTQyNzM1Ny4xNzUxMDMxNTEy_ga_HMWTBRSNTM*czE3NTEwMzE1MTEkbzEkZzEkdDE3NTEwMzE1NzIkajYwJGwwJGgw. [Accessed 10 Jan 2026].

[8] geeksforgeeks, "Functional and Non Functional Requirements," geeksforgeeks, 23 Aug 2025. [Online]. Available: https://www.geeksforgeeks.org/software-engineering/functional-vs-non-functional-requirements/. [Accessed 10 Jan 2026].

[9] A. J. B. G. R. Fulmer, "Using psychological artificial intelligence to address college student mental health," npj Digital Medicine, vol. 1, no. 1, p. 46, 2018.

[10] A. K. D. E. S. K. Lipson, "Increased rates of mental health service utilization by U.S. college students," Journal of American College Health, vol. 67, no. 7, pp. 620-627, 2019.

[11] P. S. A. D. W. S. M. Sawyer, "The age of adolescence," Lancet Child & Adolescent Health, vol. 2, no. 3, pp. 223-228, 2018.

[12] E. K. Fittes, "Student Engagement in Learning Is Low. A Survey Offers Clues on How to Bring Them Back," EdWeek Market Brief, 24 Oct 2024. [Online]. Available: https://marketbrief.edweek.org/meeting-district-needs/student-engagement-in-learning-is-low-a-survey-offers-clues-on-how-to-bring-them-back/2024/10. [Accessed 11 Jan 2026].

[13] U. G. Commission, "A Framework for Digital Transformation in State Universities of Sri Lanka," University Grants Commission, 2023.

[14] M. K. J. M. B. Bewick, "Alcohol consumption and mental health in first-year students," Studies in Higher Education, vol. 35, no. 8, pp. 899-912, 2010.

[15] JMIR Publications, "Therapeutic Potential of Social Chatbots in Alleviating Loneliness and Social Anxiety: Quasi-Experimental Mixed Methods Study," Journal of Medical Internet Research, 14 Jan 2025. [Online]. Available: https://www.jmir.org/2025/1/e65589. [Accessed 11 Jan 2026].

2. Overall Description

2.1 Product Perspective

UniPulse is conceived as a standalone, integrated platform within the educational technology landscape, designed to complement existing university support systems such as Learning Management Systems (LMS) and counseling services. Unlike traditional standalone tools, UniPulse functions as a holistic ecosystem that combines AI-driven analytics with personalized student support. The system interacts with university databases and LMS platforms to gather academic and behavioral data, while maintaining strict data privacy and ethical standards. UniPulse represents an innovative approach to student support, moving from reactive to proactive intervention models, and is positioned to set a benchmark for future AI-driven educational tools.

2.2 Product Functions

UniPulse is an AI-driven platform designed to support university freshers through four integrated core modules. The major functions of the system are:

2.2.1 Adaptive Student Profiling and Learning Persona Management

- Capture and store student demographic, academic, and initial wellbeing data during onboarding.
- Conduct weekly emotional, motivational, and social inclusion check-ins via structured forms.
- Generate and update dynamic learning personas.
- Display personalized dashboards with color-coded wellbeing indicators.
- Provide recommendations (e.g., peer matching, chatbot support) based on risk classification.

2.2.2 Social Inclusion Risk Detection and Early Intervention Management

- Collect weekly self-assessment data using Likert-scale surveys across five well-being domains.
- Apply weighted scoring (current week's highest weighted) to calculate emotional risk levels.

- Classify risk into Low, Moderate, and High categories, and detect trends (improving, worsening, etc.).
- Trigger personalized intervention recommendations based on combined risk level and trend.
- Prioritize and escalate cases requiring urgent human counselor intervention.

2.2.3 Conversational Support and Student Wellbeing Management

- Provide AI-powered conversational support for real-time emotional and academic support.
- Integrate with weekly wellbeing data to enable context-aware conversational responses.
- Implement crisis keyword detection and severity scoring (Red/Yellow/Blue flags).
- Automatically escalate high-risk conversations to human counselors with urgency.
- Log all interactions for pattern analysis and continuous system improvement.
- Provide mind relaxing activities to relax mind.

2.2.4 Intelligent Peer Matching and Relationship Evolution Management

- Administer compatibility surveys covering personality, interests, academics, and communication style.
- Calculate weighted compatibility scores across multiple dimensions (interests, academics, personality, etc.).
- Generate and suggest ranked peer matches for one-to-one or group interactions.
- Support dynamic re-matching based on weekly feedback and evolving student states.
- Enable peer connection requests, chat initiation, and study/support group formation.

2.3 User Classes and Characteristics

Since UniPulse is currently implemented as a student-only platform, students are in the primary user class.

- **User-** Student (University Fresher)

Frequency of use - Daily/Weekly

Subset of product functions used- Can access all four functions.

Technical expertise - Basic web/device literacy

Security/privilege levels- Personal login with access only to own data

Experience - Low (new to system)

2.4 Operating Environment

2.4.1 Software Requirements

- Laravel 11 (Blade + Bootstrap) – For core application development.
- MySQL – For structured storage of student, profile, and check-in data.
- Git/GitHub – For version control and collaborative development.

2.4.2 Hardware Requirements

- Minimum Intel i5/Ryzen 5 processor
- 16GB RAM
- 500GB SSD
- Linux/Windows OS

2.4.3 Database Management

UniPulse will use MySQL as the database system. Key aspects include:

- Data Security: Encryption and role-based access control.
- Reliability: Daily backups and disaster recovery mechanisms.
- Scalability: Indexed queries and optimized schemas to support large student populations.
- Compliance: GDPR-like standards to ensure ethical handling of sensitive student data.

2.4.4 Version Control

Version control will be handled using Git and GitHub, following the GitFlow branching strategy. This allows structured collaboration, pull requests, and code reviews.

2.5 Design and Implementation Constraints

- Must comply with data privacy regulations (GDPR).
- AI models must be interpretable and ethically auditable.
- Real-time chatbot responses must have <3-second latency.
- System must be accessible and mobile-responsive.
- Development must use Laravel 11 and Bootstrap for consistency and maintainability.
- Integration with future university systems (LMS, student portals) must be API-first.

2.6 Project Documentation

The project documentation contains information about the documents created throughout the project lifecycle. For UniPulse, the following key documents have been or will be produced:

Project Proposal Report- The project proposal outlines the foundational vision, objectives, and feasibility of UniPulse. It includes a detailed problem statement, proposed AI-driven solution, methodology (agile development), high-level system architecture, and initial risk assessment. The proposal also covers resource planning, timeline estimates, and ethical considerations related to student data privacy and AI ethics. This document served as the basis for project approval and stakeholder alignment.

Software Requirements Specification (SRS)- This document provides a comprehensive and structured description of UniPulse's functional and non-functional requirements. It acts as the primary reference for developers, testers, and stakeholders, ensuring a shared understanding of system capabilities, constraints, and user interactions. Key sections include overall description, system features (use case driven), external interfaces, and non-functional requirements such as performance, security, and usability.

Project Management Tool Report- This report documents the project's execution using agile project management tools (e.g., Jira). It includes sprint logs, task breakdowns, burndown charts, team collaboration metrics, and progress tracking against milestones. The report reflects how the project was managed iteratively, highlighting adjustments made based on feedback, risks encountered, and resource utilization throughout development.

User Acceptance Testing (UAT) Report- The UAT report captures the testing phase where real end-users (university freshers) validate the system against their needs. It includes test scenarios, participant feedback, defect logs, and pass/fail results for each module (profiling, peer matching, risk detection, chatbot). This document ensures the system meets user expectations and is ready for deployment, providing evidence of functionality, usability, and emotional support effectiveness.

Final Project Report- This comprehensive report summarizes the entire project lifecycle, from inception to deployment. It includes the project vision, implementation methodology, technical architecture, challenges faced, lessons learned, and outcomes achieved. The report also evaluates system performance against initial objectives, discusses scalability and ethical considerations, and provides recommendations for future enhancements, including admin dashboard development and institutional scaling.

2.7 User Documentation

Student User Manual / Guide- A detailed, student-focused manual will be provided to help users navigate and utilize all features of UniPulse. The manual includes step-by-step instructions for:

- Account creation and profile setup
- Completing weekly emotional and social check-ins
- Assessing the risk using the risk detection component
- Interacting with the AI-powered chatbot for support
- Using the peer matching system to connect with other students
- Understanding dashboard indicators and personalized recommendations

2.8 Assumptions and Dependencies

Assumptions:

- Students will consistently complete weekly emotional and social check-ins.
- No admin dashboard is required in the initial release.
- Ethical approval for data collection and AI usage will be secured before launch.

Dependencies:

- Laravel 11 and MySQL for core application functionality.
- GitHub for version control and team collaboration.
- Reliable cloud hosting for deployment.
- Compliance with institutional data governance policies.

3. External Interface Requirements

3.1 User Interfaces

Adaptive Student Profiling and Learning Persona Management

Upon successful authentication, the user is redirected to the main dashboard, as illustrated in Figure 3.1. This dashboard serves as the central interface of the system, providing users with quick access to key functionalities and system features. In addition, the student's personal and academic profile details are displayed as shown in Figure 3.2, enabling users to review and manage information efficiently.

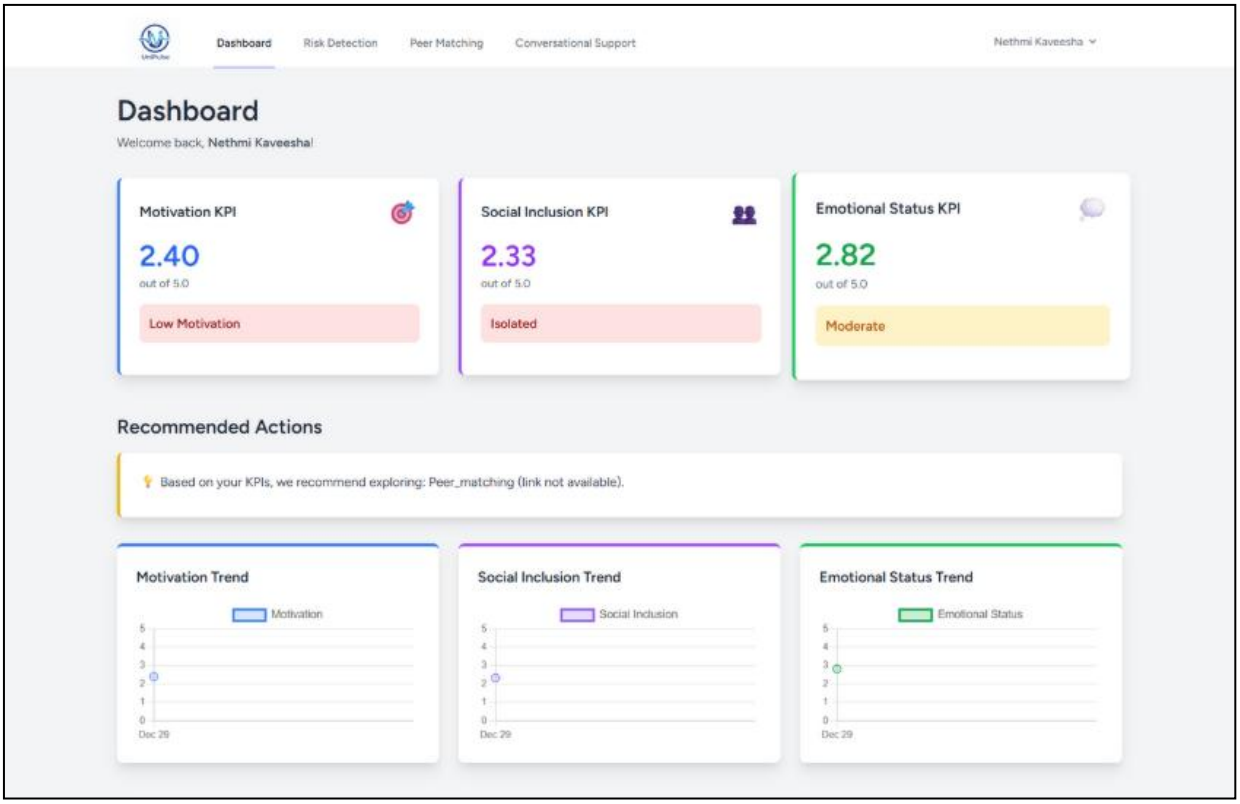


Figure 3.1: Student Dashboard

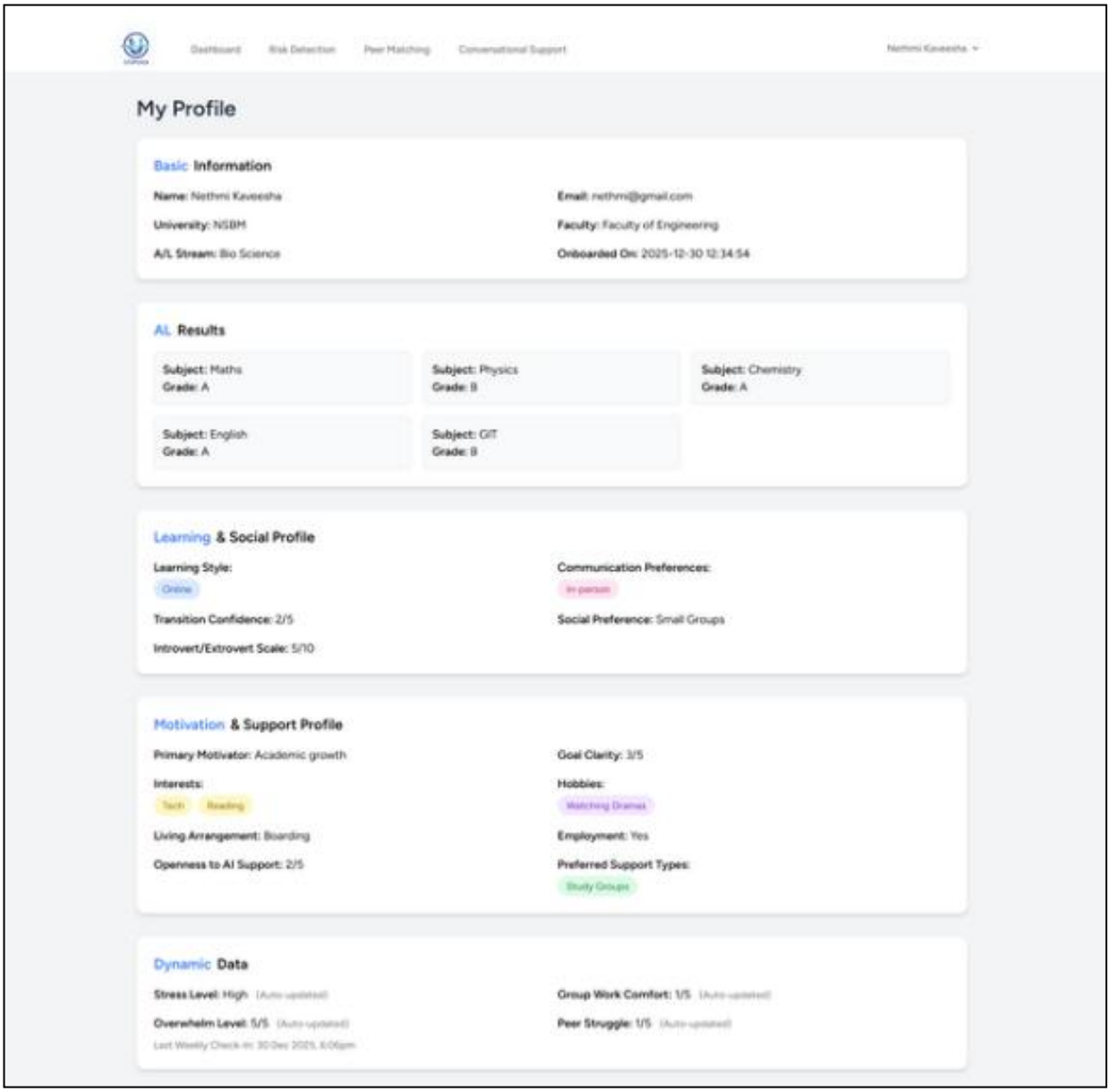


Figure 3.2: Student's Profile

Social Inclusion Risk Detection and Early Intervention Management

When the user navigates to the Risk Detection tab, the risk detection dashboard is presented, as depicted in Figure 3.3. This interface is designed to highlight potential risk indicators related to student engagement and wellbeing. Furthermore, analytical visualizations and graphical summaries are displayed, as shown in Figure 3.4, allowing users to view trends and insights in a clear and intuitive manner.

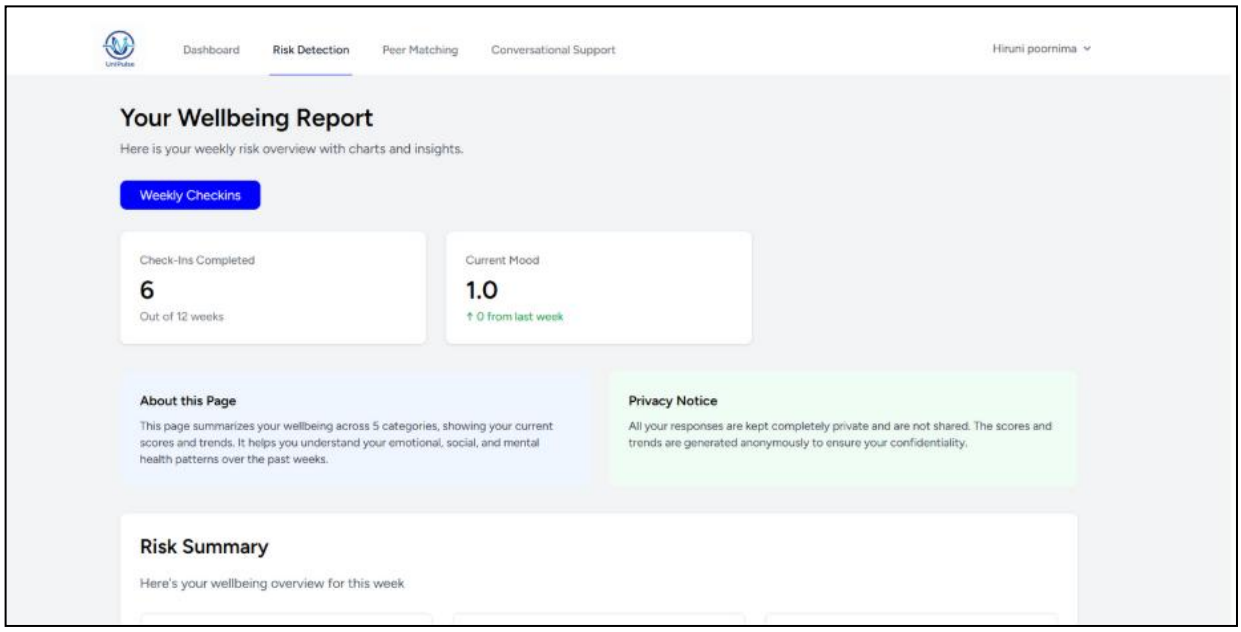


Figure 3.3: Risk Detection Dashboard

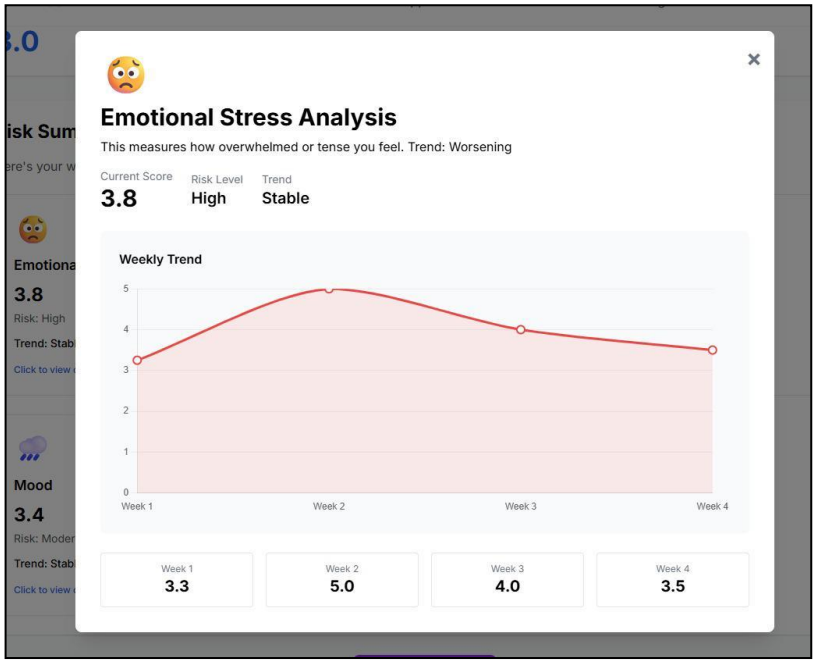


Figure 3.4: Analysis Graph

Conversational Support and Student Wellbeing Management

The main interface of the Conversational Support module is shown in Figure 3.5. This interface enables students to interact with the system through a chat-based mechanism. When the user initiates a conversation, the system generates context-aware responses, which are displayed as illustrated in Figure 3.6, thereby providing continuous guidance and emotional support.

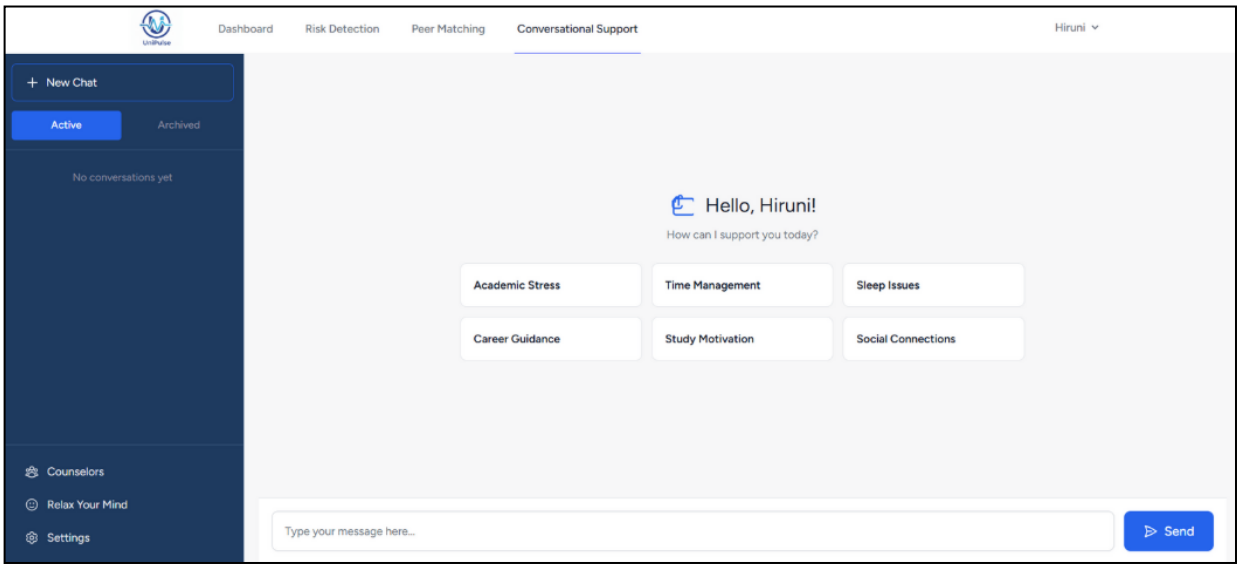


Figure 3.5: Conversational Support Chat Interface

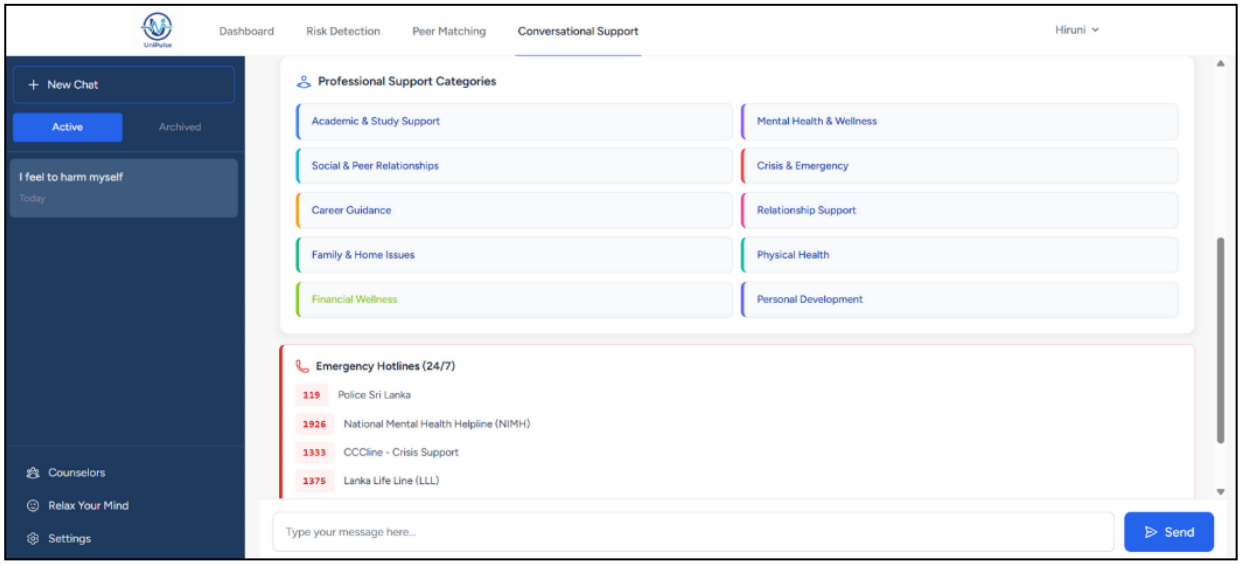


Figure 3.6: Conversational Support Chat Interface

Intelligent Peer Matching and Relationship Evolution Management

As shown in Figure 3.8, peer suggestions are presented to users along with a compatibility score, facilitating informed decision-making when forming connections. Additionally, options for group creation and peer discovery are available, as demonstrated in Figure 3.7. These features support social engagement by enabling students to connect with suitable peers and form collaborative groups based on shared interests and compatibility metrics.

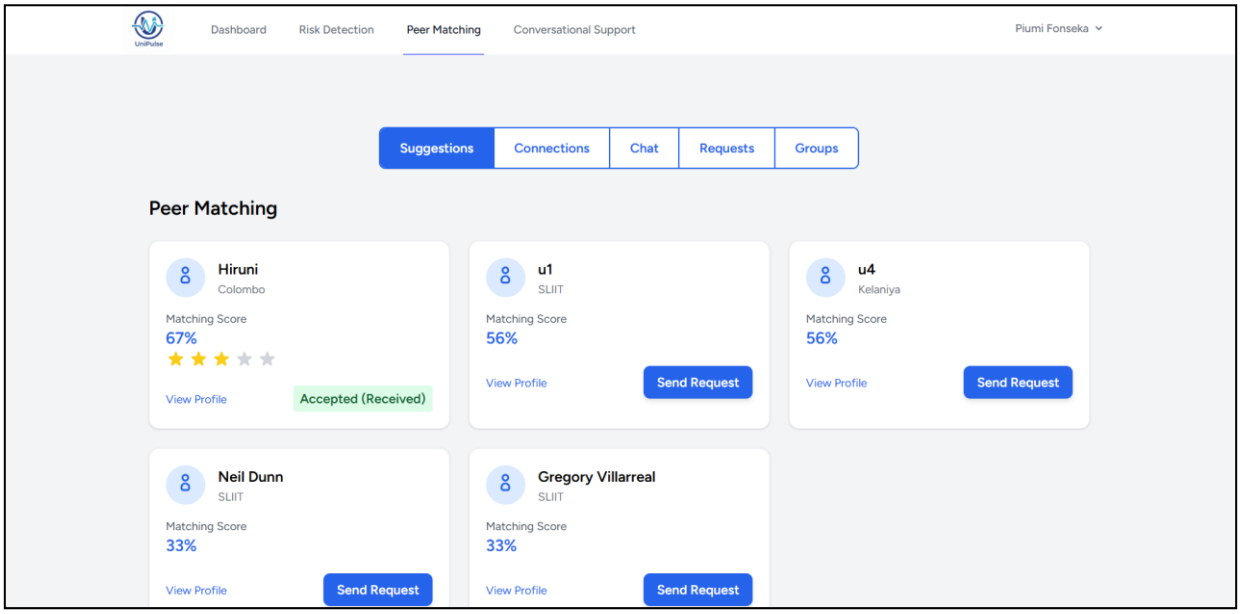


Figure 3.7: Peer Suggestions

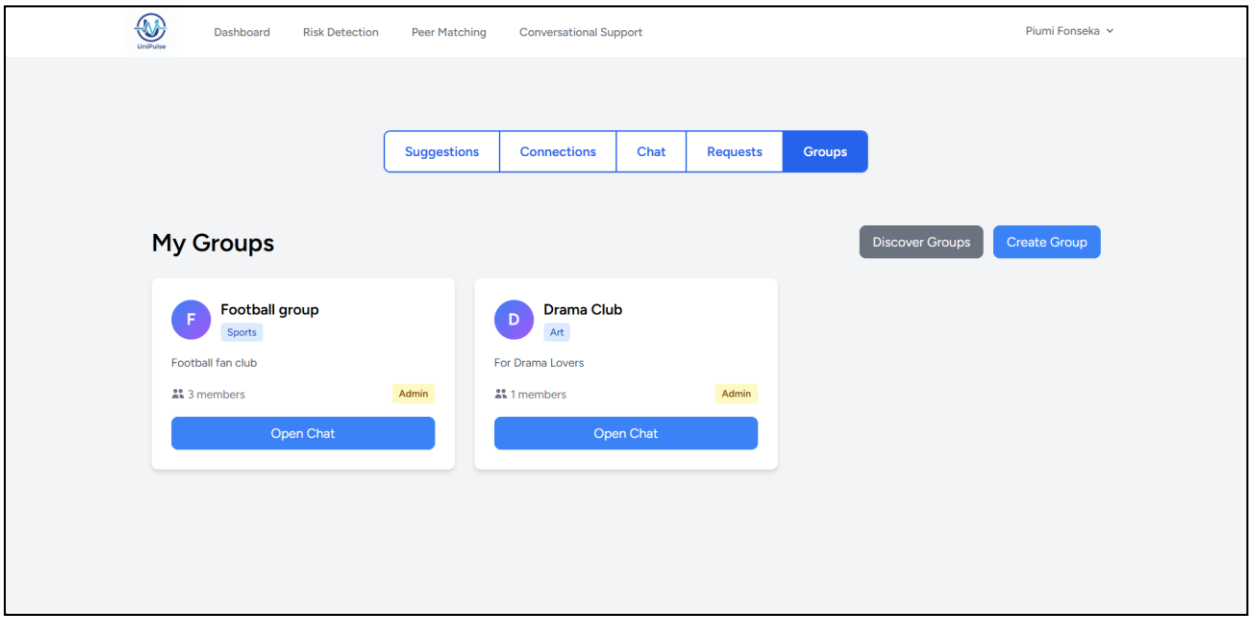


Figure 3.8: Group Formation and Discovery

3.2 Hardware Interfaces

UniPulse is a web-based platform accessible through standard computing devices such as laptops, desktops, tablets, and smartphones. No specialized hardware is required to operate the system. Users interact with the platform via input devices such as keyboards, touchscreens, and pointing devices.

The system must support the following hardware configurations:

- Desktop and laptop computers with modern web browsers
- Tablets and smartphones with touch-based interfaces
- Minimum screen resolution: 1280 × 720 for optimal UI rendering
- Internet-enabled devices with Wi-Fi or Ethernet connectivity

3.3 Software Interfaces

UniPulse interacts with multiple software components to ensure seamless functionality and data processing. The following software interfaces are required:

Operating System: Windows 10 or above

Server-Side Software: Laravel 11 (PHP framework) for backend logic, MySQL for database management

Client-Side Software: Web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest versions)

Development Tools: XAMPP (local development environment), phpMyAdmin (database administration tool), GitHub (version control and collaboration), IDE: Visual Studio Code

External System Interfaces (Future Integration):

- Learning Management Systems (LMS) via REST APIs
- University student portals
- Counseling and support systems

3.4 Communications Interfaces

UniPulse is a web platform that relies on standard internet communication protocols for data exchange and system interaction. All communication between users and the system will be encrypted using HTTPS to protect sensitive emotional and personal data. In-app messaging will be used for chatbot interactions and system notifications. The platform must support real-time communication for chatbot responses with a maximum response latency of 3 seconds.

4. System Features

Figure 4.1 depicts the use case diagram of the system. University fresher is the main actor in this system.

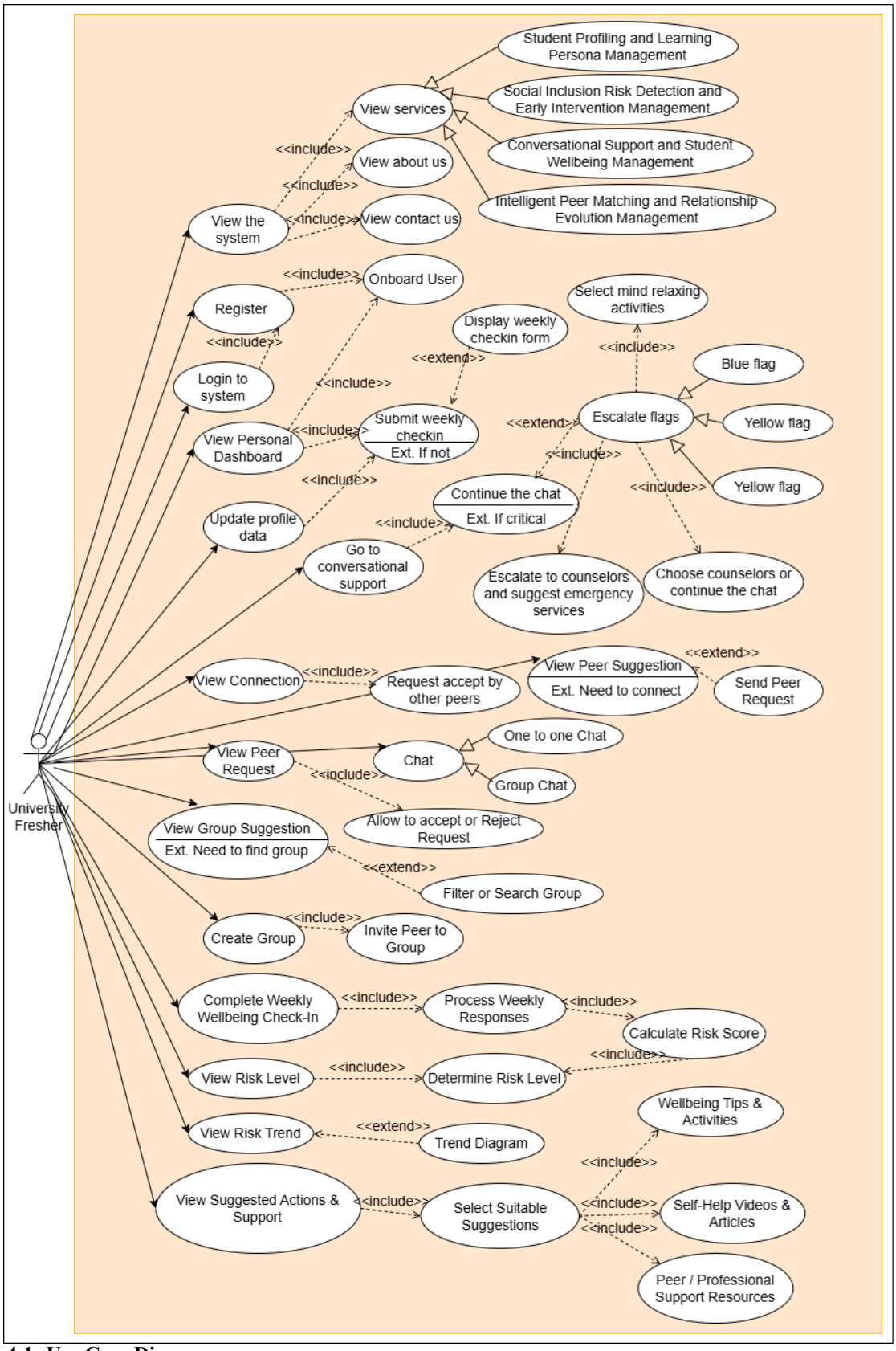


Figure 4.1: Use Case Diagram

4.1 Student Profiling and Onboarding Feature

Description

This feature enables the creation and continuous updating of student profiles. The system collects academic, personal, social, and emotional data during onboarding and through periodic interactions. The profile serves as the foundational data source for peer matching, conversational flow, and emotional risk detection.

Actors

- Student

Functional Requirements

- The system shall allow students to register using basic personal and academic information.
- The system shall collect interest, personality, and communication preference data during onboarding.
- The system shall store emotional, social inclusion and motivational data submitted through weekly check-ins.
- The system shall dynamically update student profiles based on new inputs.

Pre-conditions

- Students must be authenticated.

Post-conditions

- A comprehensive and updated student profile is available for system processing.

4.2 Emotional Risk Detection and Intervention Feature

Description

This feature monitors emotional wellbeing and engagement patterns to detect early signs of emotional risk. The system performs trend analysis and initiates low-intensity interventions without replacing human decision-making.

Actors

- Student

Functional Requirements

- The system shall analyze emotional check-in data over time.
- The system shall detect sustained stress or disengagement patterns.
- The system shall initiate preventive actions such as peer rematching or supportive prompts.
- The system shall notify authorized personnel if risk thresholds are exceeded.

Pre-conditions

- Sufficient historical emotional data must be available.

Post-conditions

- Appropriate intervention actions are executed or recommended.

4.3 Conversational Support and Interaction Feature

Description

This feature provides a guided conversational interface that supports students through structured prompts and system-driven interactions. The conversational flow adapts based on student profile changes and emotional state.

Actors

- Student

Functional Requirements

- The system shall provide conversational prompts based on student context.
- The system shall adapt responses according to emotional and engagement indicators.
- The system shall encourage peer interaction and self-reflection through dialogue.
- The system shall log conversational interactions for analysis.

Pre-conditions

- Student must be logged into the system.

Post-conditions

- Conversational data is stored and contributes to profile updates.

4.4 Intelligent Peer Matching Feature

Description

This feature provides intelligent peer recommendations by analyzing student profiles using a weighted scoring algorithm. The system evaluates compatibility across multiple dimensions and supports dynamic rematching when interaction quality declines.

Actors

- Student

Functional Requirements

- The system shall calculate peer compatibility scores using academic, social, and emotional indicators.
- The system shall display peer recommendations with explanatory attributes.
- The system shall allow students to accept or ignore suggested matches.
- The system shall recalculate compatibility scores when profile data changes.

Pre-conditions

- Student profile must be available and complete.

Post-conditions

- One or more peer connections are established or updated.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall support up to 1,000 concurrent users with an average response time of under 3 seconds for all UI interactions.
- The AI chatbot shall generate responses within 5 seconds of receiving a student message.
- Peer matching recommendations shall be generated within 3 seconds of a student's request or profile update.
- Weekly emotional check-ins and risk analysis processing shall complete within 10 seconds per student.
- The system dashboard shall load within 3 seconds, including all personalized indicators and notifications.
- Database queries for profile updates, matching, and reporting shall execute in under 1 second under normal load.
- Rationale: University freshers may use the platform during peak times (e.g., start of term, exam periods). Fast response times ensure usability and trust in the system's real-time support capabilities.

5.2 Safety Requirements

- The system shall not provide medical, psychiatric, or crisis intervention advice beyond pre-defined supportive prompts.
- In cases where a student indicates self-harm, severe distress, or imminent danger, the system shall immediately escalate the case to a human counselor by providing the student with contact information of emergency services and counselors.
- All AI-driven interventions must be reviewed and approved by a qualified mental health professional before implementation.
- Students shall be clearly informed that the system is a supportive tool and not a replacement for professional counseling.
- The system shall include clear disclaimers and guidelines for appropriate use during onboarding and within the chatbot interface.

5.3 Security Requirements

- All user data including personal, emotional, and interaction data shall be encrypted in database.
- The system shall implement role-based access control (RBAC) to ensure students can only access their own data.
- User authentication shall require strong passwords (min. 8 characters, mixed case, numbers, symbols) and support optional multi-factor authentication (MFA).
- The system shall comply with GDPR and similar data protection regulations, including provisions for data subject access, rectification, and deletion.
- All third-party integrations (e.g., LMS, cloud hosting) shall use secure API keys and OAuth 2.0 where applicable.
- Regular security audits and vulnerability assessments shall be conducted, with patches applied within 48 hours of identification.

5.4 Software Quality Attributes

Usability:

- The system shall achieve a System Usability Scale (SUS) score of at least 80 in UAT.
- All primary user tasks (profile setup, check-in, chatting, matching) shall be completable within 3 minutes by a first-time user.

Reliability:

- The system shall maintain 99.5% uptime during academic terms.
- Critical features (risk detection, chatbot, matching) shall have a mean time between failures (MTBF) of at least 720 hours.

Maintainability:

- The codebase shall maintain test coverage of at least 85% for core modules.
- All APIs and database schemas shall be fully documented following OpenAPI standards.

Scalability:

- The system shall be deployable in a containerized environment to support horizontal scaling.

- Database design shall support sharding and indexing to handle up to 1,000 students without performance degradation.

Accessibility:

- The platform shall comply with standards to ensure accessibility for users.
- All images, buttons, and form fields shall include appropriate labels and keyboard navigation support.
- The platform shall be accessible 24/7 across all major web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

5.5 Business Rules

- Only registered university freshers with a valid institutional email address may create an account.
- Students may update their own profiles but cannot modify emotional check-in data after submission.
- Peer matching suggestions shall only be generated for students who have completed their onboarding profile.
- Emotional risk alerts shall be escalated to counselors only if a student's risk level is high.
- All system-generated interventions (e.g., peer rematching, chatbot prompts) must be logged and reviewed monthly for effectiveness and ethical compliance.
- Data retention: Students may delete their profile and associated personal data at any time.

6. Other Requirements

6.1 Database Requirements

- Utilize PhpMyAdmin for managing MySQL databases.
- Ensure compatibility with MySQL databases.

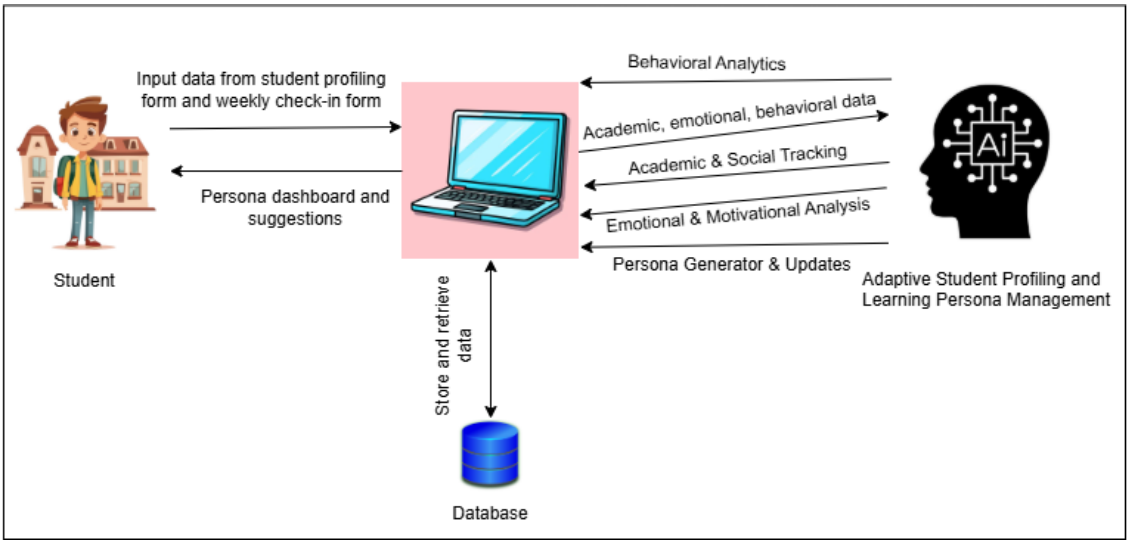
6.2 Integration Requirements

- Include required PHP libraries via Composer (e.g., Laravel, PHPUnit).
- Use XAMPP for local development (Apache, MySQL, PHP).
- Ensure compatibility with XAMPP's PHP and MySQL configuration.
- Set up Git/GitHub integration within Eclipse for version control.

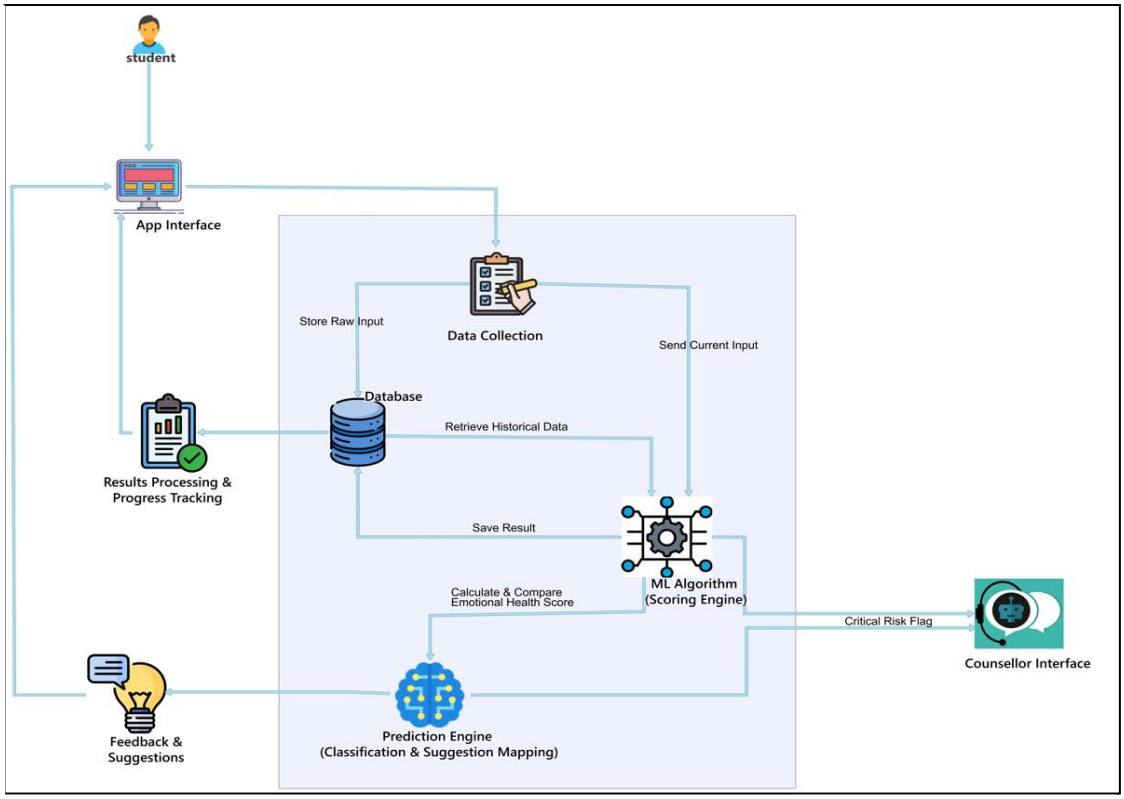
Appendix A: Glossary

- SRS: Software Requirements Specification
- AI: Artificial Intelligence
- GUI: Graphical User Interface
- MySQL: Relational Database Management System
- PHP: Hypertext Preprocessor
- Laravel: PHP-based Web Application Framework
- Git: Version Control System
- GitHub: Online Code Hosting and Collaboration Platform
- API: Application Programming Interface
- LMS: Learning Management System
- UAT: User Acceptance Testing
- GDPR: General Data Protection Regulation
- HTTP: Hypertext Transfer Protocol
- UML: Unified Modeling Language

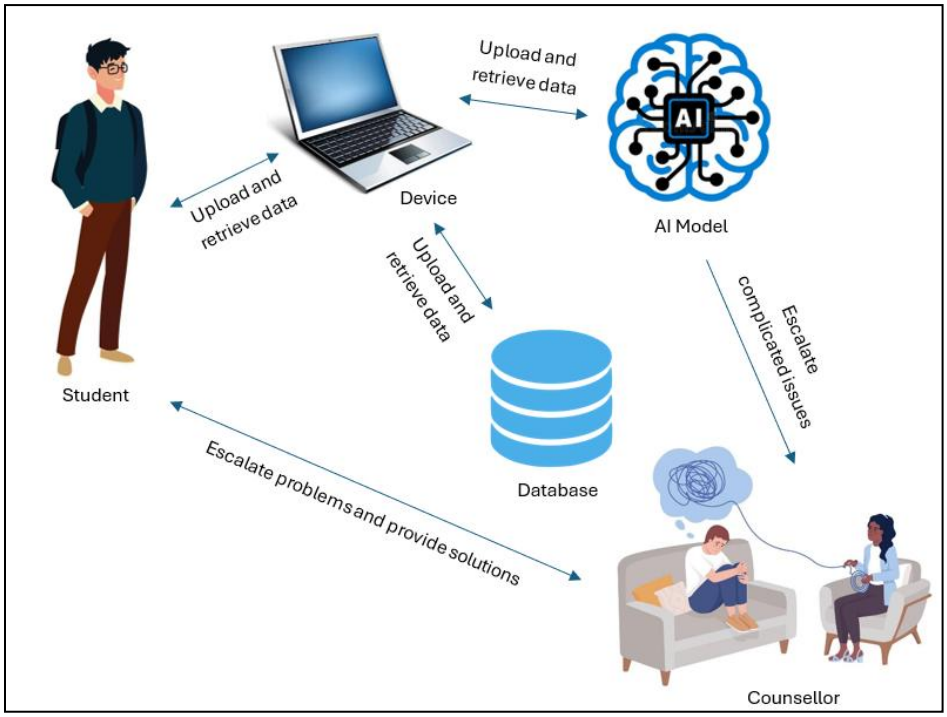
Appendix B: Analysis Models



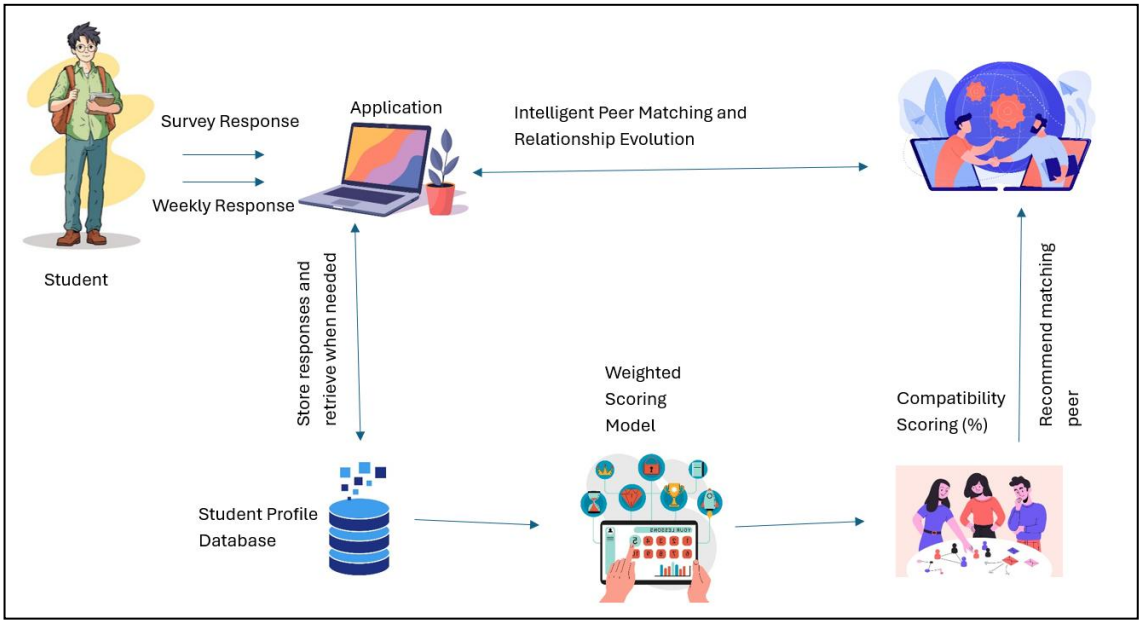
Appendix B.1: System Architecture Diagram for Adaptive Student Profiling



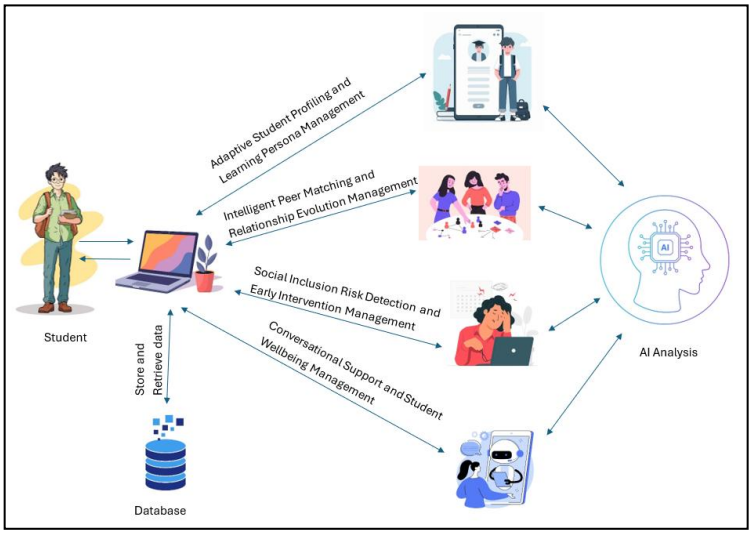
Appendix B.2: System Architecture Diagram for Risk Detection



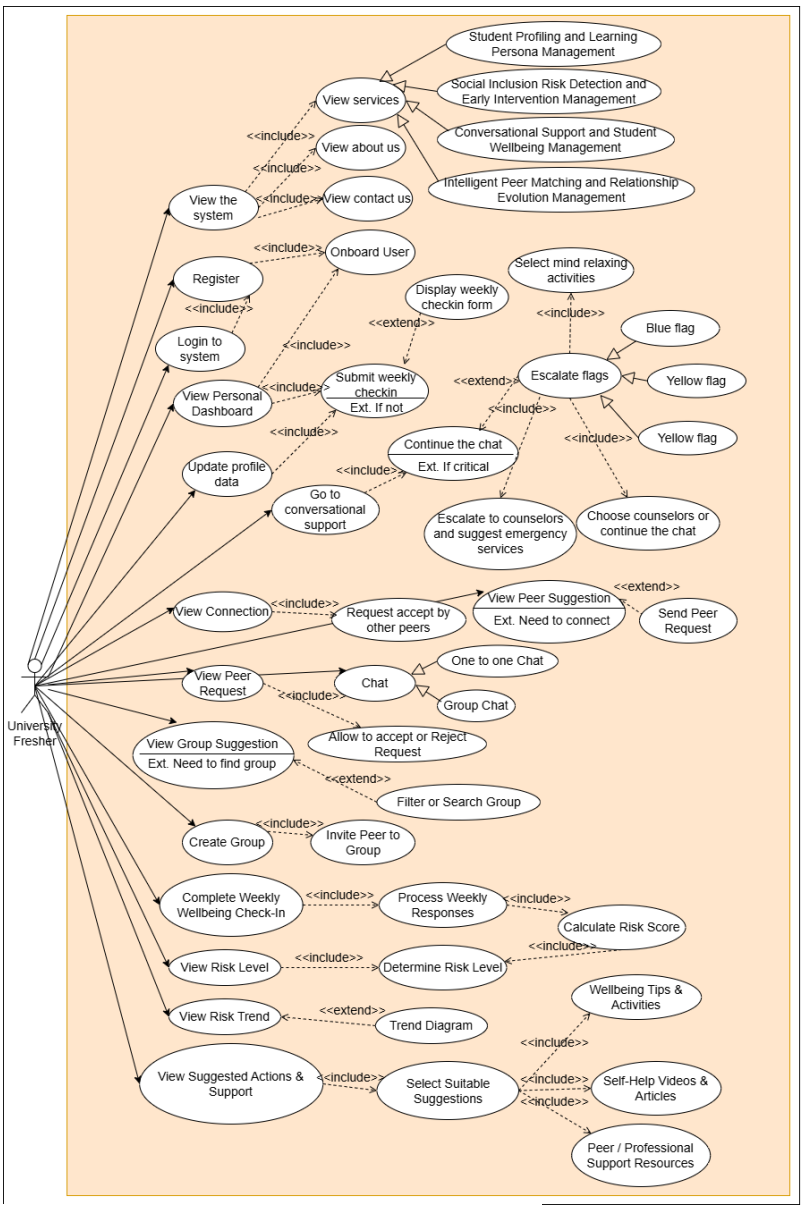
Appendix B.3: System Architecture Diagram for Conversational Support



Appendix B.4: System Architecture Diagram for Peer Matching



Appendix B.5: Overall System Architecture



Appendix B.6: Use Case Diagram

Appendix C: To Be Determined List

TBD-1: Peer Matching Weight Configuration

Identification of the final weight distribution for academic, social, personality, and emotional attributes used in the peer matching algorithm.

TBD-2: Emotional Risk Threshold Definition

Definition of threshold values and conditions required to classify emotional risk levels for students.

TBD-3: Conversational Flow Rules

Identification of conversational prompt rules, response variations, and frequency of system-initiated interactions.

TBD-4: Profiling Data Attributes

Finalization of additional data attributes required for adaptive student profiling.

TBD-5: Data Retention and Privacy Policies

Definition of data retention periods and privacy handling procedures for student interaction and emotional data.